

Sydney Girls High School



2008 HSC Assessment Task 2

MATHEMATICS

Extension 2

Time Allowed: 90 minutes

Topic: Complex Numbers

Directions to Candidates:

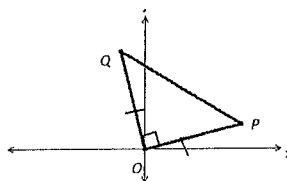
- There are THREE (3) questions, of equal value.
- All questions must be attempted.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Write on one side of the paper only.
- Diagrams are NOT to scale.

Question 1: (25 marks)

- a) If $z = 3 - 2i$ and $\omega = 1 + 4i$, find in the form $x + iy$:
- | | | |
|------|----------------------|---|
| i. | $2z + 3\omega$ | 2 |
| ii. | $iz - \omega$ | 2 |
| iii. | $\frac{\omega}{z}$ | 2 |
| iv. | $\overline{z\omega}$ | 3 |
- b)
- | | | |
|-----|---|---|
| i. | Find $\sqrt{-3 - 4i}$ and express each answer in the form $x + iy$. | 4 |
| ii. | Using (i) or otherwise, solve the equation $z^2 - 3z + (3 + i) = 0$. | 2 |
- c)
- | | | |
|-----|--|---|
| i. | Express $z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$ in mod-arg form. | 2 |
| ii. | Hence express z^6 in the form $x + iy$, where x and y are real numbers. | 2 |
- d)
- | | | |
|-----|---|---|
| i. | If $z = \cos\theta + i\sin\theta$, show that $z^n + \frac{1}{z^n} = 2\cos n\theta$ | 2 |
| ii. | Hence or otherwise express $\cos^4\theta$ in terms of multiples of θ . | 4 |

Question 2: (25 marks)

- a) The points P and Q in the complex plane correspond to the complex numbers z and w respectively. The triangle OPQ is isosceles and $\angle POQ$ is a right angle. Show that $z^2 + w^2 = 0$.



- b) Sketch the region in the complex plane where the inequalities $|z+1-2i| \leq 3$ and $-\frac{\pi}{3} \leq \arg z \leq \frac{\pi}{4}$ both hold. 3
- c) Sketch the locus of the following. Draw separate diagrams. 2
- $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{3}$ 2
 - $\operatorname{Re}(z) + \operatorname{Im}(z) = 3$ 2
- d) i. Sketch the locus of $|z-1+2i| = |z+3|$. 2
- ii. Find the locus of z . 2
- e) Find the locus of z if $w = \frac{z-2i}{z+2}$ is purely imaginary. 3
- f) i. If $z = x+iy$, sketch on an Argand diagram, the curve defined by the equation $\operatorname{Im}(z-2+i) = 3$. 2
- ii. Find, using your diagram, the minimum value of $|z|$ subject to this condition. 1

- g) i. Prove that $|z|^2 = z\bar{z}$ 1
- ii. Prove that for all complex numbers z and w : 2

$$|z+w|^2 + |z-w|^2 = 2(|z|^2 + |w|^2)$$

- h) The point A on an Argand diagram represents the complex number $1+i$. 3
- Find the complex number represented by B if OBA is an equilateral triangle and B is in the second quadrant.

Question 3: (25 marks)

- a) i. Find in mod-arg form the five roots of $z^5 = -1$. 2
- ii. Hence factorise $z^5 + 1$ over the real field. 3
- iii. Show that $z^4 - z^3 + z^2 - z + 1 = \left(z^2 - 2z \cos \frac{\pi}{5} + 1\right) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1\right)$. 3
- iv. Show that $\cos \frac{3\pi}{5} + \cos \frac{\pi}{5} = \frac{1}{2}$ and $\cos \frac{3\pi}{5} \cos \frac{\pi}{5} = -\frac{1}{4}$. 2
- v. Deduce that $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$ are the roots of the equation $4x^2 - 2x - 1 = 0$. 2
- vi. Find the exact values of $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$. 2
- b) ω is the complex root of $z^6 - 1 = 0$ with the smallest positive argument.
- Find the two real roots of $z^6 - 1 = 0$. 1
 - Prove that ω , ω^2 , ω^4 and ω^5 are the roots of $z^4 + z^2 + 1 = 0$. 2
 - Find the quadratic equation whose roots are $\alpha = \omega + \omega^5$ and $\beta = \omega^2 + \omega^4$. 3
- c) i. Express $(3+2i)(5+4i)$ and $(3-2i)(5-4i)$ in the form $a+ib$. 2
- ii. Hence find the prime factors of $7^2 + 22^2$. 3

2008 EXTENSION 2 MATHEMATICS – HSC ASSESSMENT TASK 2 SOLUTIONS

Question 1:

a) $z = 3 - 2i$ and $w = 1 + 4i$

i. $2z + 3w = 2(3 - 2i) + 3(1 + 4i)$
 $= 6 - 4i + 3 + 12i$
 $= 9 + 8i$

ii. $iz - w = i(3 - 2i) - (1 + 4i)$
 $= 3i - 2i^2 - 1 - 4i$
 $= 3i + 2 - 1 - 4i$
 $= 1 - i$

iii. $\frac{w}{z} = \frac{1 + 4i}{3 - 2i} \times \frac{3 + 2i}{3 + 2i}$
 $= \frac{(1 + 4i)(3 + 2i)}{(3 - 2i)(3 + 2i)}$
 $= \frac{3 + 14i + 8i^2}{9 - 4i^2}$
 $= \frac{3 + 14i - 8}{9 + 4}$
 $= \frac{-5 + 14i}{13}$

iv. $\overline{zw} = \overline{z} \times \overline{w}$
 $= (3 + 2i)(1 - 4i)$
 $= 3 - 10i - 8i^2$
 $= 3 - 10i + 8$
 $= 11 - 10i$

b) Let $x + iy = \sqrt{-3 - 4i}$

i. $(x + iy)^2 = -3 - 4i$
 $x^2 + i2xy + i^2y^2 = -3 - 4i$
 $(x^2 - y^2) + i2xy = -3 - 4i$

Equating real and imag.
coefficients:

$x^2 - y^2 = -3$ ---- (1)
 $2xy = -4$ ---- (2)

$(x^2 + y^2)^2 = (x^2 - y^2)^2 + (2xy)^2$
 $(x^2 + y^2)^2 = (-3)^2 + (-4)^2$
 $= 25$

$x^2 + y^2 = 5$ ---- (3)

(1) + (3):

$2x^2 = 2$

$x^2 = 1$

$x = \pm 1$

From (2):

When $x = 1, y = -2$

When $x = -1, y = 2$

$\therefore \sqrt{-3 - 4i} = \pm(1 - 2i)$

ii. $z^2 - 3z + (3 + i) = 0$

$z = \frac{3 \pm \sqrt{9 - 4(3 + i)}}{2}$

$= \frac{3 \pm \sqrt{-3 - 4i}}{2}$

$= \frac{3 \pm (1 - 2i)}{2}$

$z = \frac{3 - (1 - 2i)}{2} \quad z = \frac{3 + (1 - 2i)}{2}$

$= \frac{2 + 2i}{2} \quad = \frac{4 - 2i}{2}$

$= 1 + i \quad \text{or} \quad = 2 - i$

c) $z = \frac{1}{2} + \frac{\sqrt{3}}{2}i$

i. $|z| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

$= \sqrt{\frac{1}{4} + \frac{3}{4}}$

$= 1$

$\arg z = \tan^{-1} \frac{y}{x}$
 $\theta = \tan^{-1} \left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}} \right)$
 $= \tan^{-1} \sqrt{3}$
 $= \frac{\pi}{3}$

$\therefore z = \text{cis} \frac{\pi}{3}$

ii. $z^6 = \left(\text{cis} \frac{\pi}{3} \right)^6$
 $= \text{cis} \frac{6\pi}{3}$
 $= \text{cis} 2\pi$
 $= \cos 2\pi + i \sin 2\pi$
 $= 1$

d)

i. $z = \cos \theta + i \sin \theta$
 $z^n + \frac{1}{z^n} = z^n + z^{-n}$
 $= (\cos \theta + i \sin \theta)^n + (\cos \theta + i \sin \theta)^{-n}$
 $= \cos n\theta + i \sin n\theta + \cos(-n\theta) + i \sin(-n\theta)$
 $= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$
 $= 2 \cos n\theta$

ii. $z + \frac{1}{z} = 2 \cos \theta$
 $\left(z + \frac{1}{z} \right)^4 = (2 \cos \theta)^4$
 $= 16 \cos^4 \theta$
 $\left(z + \frac{1}{z} \right)^4 = z^4 + 4z^2 + 6 + \frac{4}{z^2} + \frac{1}{z^4}$
 $16 \cos^4 \theta = \left(z^4 + \frac{1}{z^4} \right) + 4 \left(z^2 + \frac{1}{z^2} \right) + 6$
 $= 2 \cos 4\theta + 8 \cos 2\theta + 6$
 $\cos^4 \theta = \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}$

Question 2:

a)

Transformation from OP to OQ is a rotation of $+90^\circ$.

$w = iz$

$w^2 = i^2 z^2$

$= -z^2$

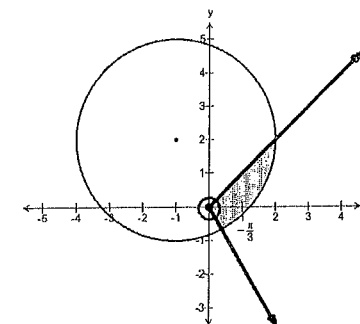
$z^2 + w^2 = 0$

b)

$|z + 1 - 2i| \leq 3$

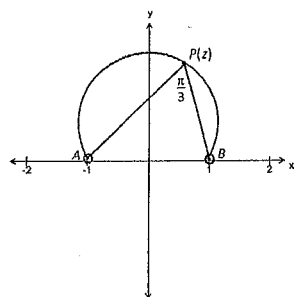
$|z - (-1 + 2i)| \leq 3$

Circle centre $(-1, 2)$ radius 3 units.



c)

i.



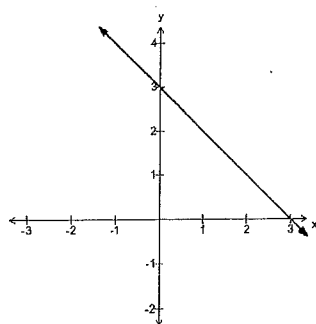
Locus of z is the major arc of a circle excluding A and B .

ii. If $z = x + iy$, then:

$$\operatorname{Re}(z) = x$$

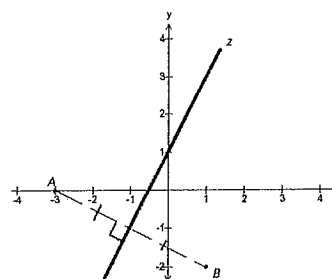
$$\operatorname{Im}(z) = y$$

$$\therefore \text{locus of } z \text{ is } x + y = 3$$



d)

i.



ii. locus of z is the perpendicular bisector of AB .

Midpoint AB :

$$M_{AB} = \left(\frac{-3+1}{2}, \frac{0-0}{2} \right) = (-1, 0)$$

Gradient AB :

$$m_{AB} = \frac{-2-0}{1+3} = -\frac{1}{2}$$

So gradient of locus of z is 2

Equation of locus of z :

$$y + 1 = 2(x + 1)$$

$$y + 1 = 2x + 2$$

$$2x - y + 1 = 0$$

e) Let $z = x + iy$

Algebraically:

$$w = \frac{x + iy - 2i}{x + iy + 2}$$

$$= \frac{x + i(y-2)}{(x+2) + iy} \times \frac{(x+2) - iy}{(x+2) - iy}$$

$$= \frac{x(x+2) - ixy + i(y-2)(x+2) - i^2y(y-2)}{[(x+2) + iy][(x+2) - iy]}$$

$$w = \frac{x(x+2) - ixy + i(y-2)(x+2) - i^2y(y-2)}{[(x+2) + iy][(x+2) - iy]} \\ = \frac{x^2 + 2x - ixy + i(xy + 2y - 2x - 4) + y^2 - 2y}{(x+2)^2 - i^2y^2} \\ = \frac{x^2 + 2x + y^2 - 2y}{(x+2)^2 + y^2} + i \frac{2y - 2x - 4}{(x+2)^2 + y^2}$$

If w is purely imaginary, then $\operatorname{Re}(z) = 0$.

$$\frac{x^2 + 2x + y^2 - 2y}{(x+2)^2 + y^2} = 0$$

$$x^2 + 2x + y^2 - 2y = 0$$

$$x^2 + 2x + 1 + y^2 - 2y + 1 = 2$$

$$(x+1)^2 + (y-1)^2 = 2$$

Locus of z is a circle, centre $(-1, 1)$ and radius $\sqrt{2}$ units, excluding $(0, 2)$ and $(-2, 0)$.

Geometrically:

If w is purely imaginary then $\arg w = \pm \frac{\pi}{2}$.

$$w = \frac{z - 2i}{z + 2}$$

$$\arg w = \arg(z - 2i) - \arg(z + 2)$$

$$\pm \frac{\pi}{2} = \arg(z - 2i) - \arg(z + 2)$$

Locus of z is a circle with $(0, 2)$ and $(-2, 0)$ as endpoints of the diameter.

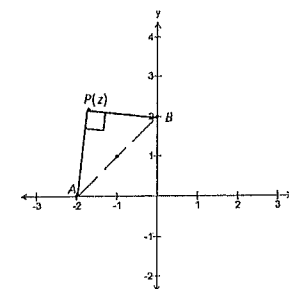
Centre of circle:

$$M_{AB} = \left(\frac{-2+0}{2}, \frac{0+2}{2} \right) = (-1, 1)$$

Diameter $= 2\sqrt{2}$ (using Pythagoras' Thm)

Radius $= \sqrt{2}$

Locus of z is a circle, centre $(-1, 1)$ and radius 2 units, excluding $(0, 2)$ and $(-2, 0)$.



f) If $z = x + iy$, then:

$$\operatorname{Im}(z - 2 + i) = 3$$

$$\operatorname{Im}(x + iy - 2 + i) = 3$$

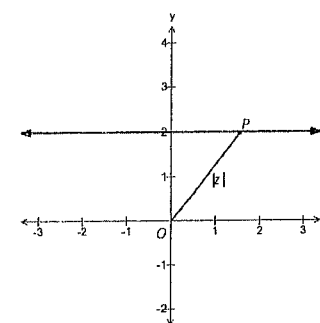
$$\operatorname{Im}(\{x - 2\} + i\{y + 1\}) = 3$$

$$y + 1 = 3$$

$$y = 2$$

$$y + 1 = 3$$

$$y = 2$$



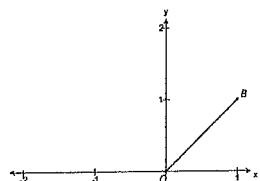
If P represents z , then $OP = |z|$. When P has coordinates $(0, 2)$, minimum $|z| = 2$.

g)

$$\begin{aligned}
 \text{i. } LHS &= |z|^2 \\
 &= (\sqrt{x^2 + y^2})^2 \\
 &= x^2 + y^2 \\
 &= x^2 - i^2 y^2 \\
 &= (x + iy)(x - iy) \\
 &= z\bar{z} \\
 &= RHS
 \end{aligned}$$

$$\begin{aligned}
 \text{ii. } LHS &= |z+w|^2 + |z-w|^2 \\
 &= (z+w)(\bar{z}+\bar{w}) + (z-w)(\bar{z}-\bar{w}) \\
 &= (z+w)(\bar{z}+\bar{w}) + (z-w)(\bar{z}-\bar{w}) \\
 &= \bar{z}z + \bar{z}w + w\bar{z} + w\bar{w} + \bar{z}z - \bar{z}w - w\bar{z} + w\bar{w} \\
 &= 2\bar{z}z + 2w\bar{w} \\
 &= 2(z\bar{z} + w\bar{w}) \\
 &= 2(|z|^2 + |w|^2) \\
 &= RHS
 \end{aligned}$$

h)



Transformation from OA to OB is a rotation of $+60^\circ$.

$$\begin{aligned}
 &(1+i)cis\frac{\pi}{3} \\
 &= (1+i)\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right) \\
 &= (1+i)\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{1}{2} + i\frac{\sqrt{3}}{2} + i\frac{1}{2} + i^2\frac{\sqrt{3}}{2} \\
 &= \frac{1-\sqrt{3}}{2} + i\frac{\sqrt{3}+1}{2}
 \end{aligned}$$

Question 3:

a)

$$\begin{aligned}
 \text{i. } z^5 &= -1 \\
 &= cis\pi \\
 z &= (cis\pi)^{\frac{1}{5}} \\
 &= cis\left(\frac{\pi + 2k\pi}{5}\right) \text{ where } k = 0, 1, 2, 3, 4
 \end{aligned}$$

When $k=0$: $z_0 = cis\frac{\pi}{5}$

When $k=1$: $z_1 = cis\frac{3\pi}{5}$

When $k=2$: $z_2 = cis\frac{5\pi}{5}$
 $= cis\pi$
 $= -1$

When $k=3$: $z_3 = cis\frac{7\pi}{5}$
 $= cis\left(-\frac{3\pi}{5}\right)$
 $= \bar{z}_1$

When $k=4$: $z_4 = cis\frac{9\pi}{5}$
 $= cis\left(-\frac{\pi}{5}\right)$
 $= \bar{z}_0$

QUESTION 3 CONTINUES OVERLEAF...

ii.

$$\begin{aligned}
 z^5 + 1 &= (z+1)(z-z_0)(z-\bar{z}_0)(z-z_1)(z-\bar{z}_1) \\
 &= (z+1)(z^2 - z\bar{z}_0 - z\bar{z}_0 + z_0\bar{z}_0)(z^2 - z\bar{z}_1 - z\bar{z}_1 + z_1\bar{z}_1) \\
 &= (z+1)(z^2 - \{z_0 + \bar{z}_0\}z + z_0\bar{z}_0)(z^2 - \{z_1 + \bar{z}_1\}z + z_1\bar{z}_1)
 \end{aligned}$$

As $z_0 + \bar{z}_0 = 2\text{Re}(z_0)$ and $z_1 + \bar{z}_1 = 2\text{Re}(z_1)$

$$\begin{aligned}
 &= 2\cos\frac{\pi}{5} &= 2\cos\frac{3\pi}{5} \\
 z_0\bar{z}_0 &= |z_0|^2 & z_1\bar{z}_1 &= |z_1|^2 \\
 &= 1 & &= 1
 \end{aligned}$$

$$\begin{aligned}
 z^5 + 1 &= (z+1)(z^2 - \{z_0 + \bar{z}_0\}z + z_0\bar{z}_0)(z^2 - \{z_1 + \bar{z}_1\}z + z_1\bar{z}_1) \\
 &= (z+1)\left(z^2 - 2z\cos\frac{\pi}{5} + 1\right)\left(z^2 - 2z\cos\frac{3\pi}{5} + 1\right)
 \end{aligned}$$

iii.

$$z^4 - z^3 + z^2 - z + 1 = 1 - z + z^2 - z^3 + z^4$$

G.P with $a=1$, $r=-z$, $n=5$

$$\begin{aligned}
 1 - z + z^2 - z^3 + z^4 &= \frac{1(1 - \{-z\}^5)}{1 - \{-z\}} \\
 &= \frac{1+z^5}{1+z} \\
 &= \frac{(z+1)\left(z^2 - 2z\cos\frac{\pi}{5} + 1\right)\left(z^2 - 2z\cos\frac{3\pi}{5} + 1\right)}{1+z} \text{ from (ii)} \\
 &= \left(z^2 - 2z\cos\frac{\pi}{5} + 1\right)\left(z^2 - 2z\cos\frac{3\pi}{5} + 1\right)
 \end{aligned}$$

iv.

$$\begin{aligned}
 1 - z + z^2 - z^3 + z^4 &= \left(z^2 - 2z \cos \frac{\pi}{5} + 1 \right) \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right) \\
 &= z^2 \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right) - 2z \cos \frac{\pi}{5} \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right) + 1 \left(z^2 - 2z \cos \frac{3\pi}{5} + 1 \right) \\
 &= z^4 - 2z^3 \cos \frac{3\pi}{5} + z^2 - 2z^3 \cos \frac{\pi}{5} + 4z^2 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} - 2z \cos \frac{\pi}{5} + z^2 - 2z \cos \frac{3\pi}{5} + 1 \\
 &= z^4 - 2z^3 \left(\cos \frac{3\pi}{5} + \cos \frac{\pi}{5} \right) + z^2 \left(4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} + 2 \right) - 2z \left(\cos \frac{3\pi}{5} + \cos \frac{\pi}{5} \right) + 1
 \end{aligned}$$

Equating coefficients of z^3 :

$$\begin{aligned}
 2 \left(\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} \right) &= 1 \\
 \cos \frac{\pi}{5} + \cos \frac{3\pi}{5} &= \frac{1}{2}
 \end{aligned}$$

Equating coefficients of z^2 :

$$\begin{aligned}
 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} + 2 &= 1 \\
 4 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} &= -1 \\
 \cos \frac{\pi}{5} \cos \frac{3\pi}{5} &= -\frac{1}{4}
 \end{aligned}$$

v. Let $\alpha = \cos \frac{\pi}{5}$ and $\beta = \cos \frac{3\pi}{5}$

$$\alpha + \beta = \frac{1}{2} \text{ and } \alpha\beta = -\frac{1}{4}$$

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0$$

$$x^2 - \frac{1}{2}x - \frac{1}{4} = 0$$

$$4x^2 - 2x - 1 = 0$$

$$\therefore \cos \frac{\pi}{5} \text{ and } \cos \frac{3\pi}{5} \text{ are roots of the equation } 4x^2 - 2x - 1 = 0$$

vi. $4x^2 - 2x + 1 = 0$

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{2 \pm \sqrt{4 - 4 \times 4 \times (-1)}}{2 \times 4} \\
 &= \frac{2 \pm \sqrt{20}}{8} \\
 &= \frac{2 \pm 2\sqrt{5}}{8} \\
 &= \frac{1 \pm \sqrt{5}}{4}
 \end{aligned}$$

$$\cos \frac{\pi}{5} = \frac{1 + \sqrt{5}}{4} \left(\text{as } \frac{\pi}{5} \text{ is in 1st quad.} \right) \text{ and } \cos \frac{3\pi}{5} = \frac{1 - \sqrt{5}}{4} \left(\text{as } \frac{3\pi}{5} \text{ is in 2nd quad.} \right)$$

b)

ii.

$$\begin{aligned}
 z^6 - 1 &= (z^2)^3 - 1^3 \\
 &= (z^2 - 1)(z^4 + z^2 + 1)
 \end{aligned}$$

When $z^6 - 1 = 0$:

$$(z^2 - 1)(z^4 + z^2 + 1) = 0$$

Real roots are ± 1 .

iii.

$$z^6 - 1 = 0$$

$$z^6 = 1$$

$$= \text{cis } 0$$

$$z = \text{cis} \frac{k\pi}{3} \text{ where } k = 0, 1, 2, 3, 4, 5$$

$$\text{When } k=0: \quad z_0 = \text{cis } 0 = 1$$

$$\text{When } k=2: \quad z_2 = \text{cis} \frac{2\pi}{3}$$

$$\text{When } k=4: \quad z_4 = \text{cis} \frac{4\pi}{3} = \text{cis} \left(-\frac{2\pi}{3} \right) = \overline{z_2}$$

$$\text{When } k=1: \quad z_1 = \text{cis} \frac{\pi}{3}$$

$$\begin{aligned}
 \text{When } k=3: \quad z_3 &= \text{cis} \frac{3\pi}{3} \\
 &= \cos \pi + i \sin \pi \\
 &= -1
 \end{aligned}$$

$$\text{When } k=5: \quad z_5 = \text{cis} \frac{5\pi}{3} = \text{cis} \left(-\frac{\pi}{3} \right) = \overline{z_1}$$

$$z^6 - 1 = (z-1)(z+1)(z-z_1)(z-z_2)(z-z_4)(z-z_5)$$

$$(z^2 - 1)(z^4 + z^2 + 1) = (z^2 - 1)(z-z_1)(z-z_2)(z-z_4)(z-z_5)$$

$$z^4 + z^2 + 1 = (z-z_1)(z-z_2)(z-z_4)(z-z_5)$$

Roots of $z^4 + z^2 + 1 = 0$ are z_1, z_2, z_4 and z_5

Let $\omega = cis \frac{\pi}{3}$ (complex root with the smallest positive argument)

$\omega = \left(cis \frac{\pi}{3}\right)$	$\omega^2 = \left(cis \frac{\pi}{3}\right)^2$	$\omega^4 = \left(cis \frac{\pi}{3}\right)^4$	$\omega^5 = \left(cis \frac{\pi}{3}\right)^5$
$= cis \frac{\pi}{3}$	$= cis \frac{2\pi}{3}$	$= cis \frac{4\pi}{3}$	$= cis \frac{5\pi}{3}$
$= z_1$	$= z_2$	$= cis \left(-\frac{2\pi}{3}\right)$	$= cis \left(-\frac{\pi}{3}\right)$
		$= \overline{z_2}$	$= \overline{z_1}$

So $\omega, \omega^2, \omega^4$ and ω^5 are the roots of $z^4 + z^2 + 1 = 0$.

iv.

$$\begin{aligned}\alpha &= \omega + \omega^5 \\ &= z_1 + \overline{z_1} \\ &= 2\operatorname{Re}(z_1) \\ &= 2\cos \frac{\pi}{3} \\ &= 2 \times \frac{1}{2} \\ &= 1\end{aligned}$$

$$\begin{aligned}\beta &= \omega^2 + \omega^4 \\ &= z_2 + \overline{z_2} \\ &= 2\operatorname{Re}(z_2) \\ &= 2\cos \frac{2\pi}{3} \\ &= 2 \times \left(-\frac{1}{2}\right) \\ &= -1\end{aligned}$$

$$\begin{aligned}\text{So } \alpha + \beta &= 1 + (-1) \\ &= 0\end{aligned}$$

$$\begin{aligned}\alpha\beta &= 1 \times (-1) \\ &= -1\end{aligned}$$

$$\begin{aligned}x^2 - (\alpha + \beta)x + \alpha\beta &= 0 \\ x^2 - 1 &= 0\end{aligned}$$

c)

i.

$$\begin{aligned}(3+2i)(5+4i) &= 15 + 22i + 8i^2 \\ &= 15 + 22i - 8 \\ &= 7 + 22i\end{aligned}$$

$$\begin{aligned}(3-2i)(5-4i) &= 15 - 22i + 8i^2 \\ &= 15 - 22i - 8 \\ &= 7 - 22i\end{aligned}$$

ii.

$$\begin{aligned}(7+22i)(7-22i) &= (3+2i)(5+4i)(3-2i)(5-4i) \\ 7^2 - 22^2 i^2 &= (3+2i)(3-2i)(5+4i)(5-4i) \\ 7^2 + 22^2 &= (9-4i^2)(25-16i^2) \\ &= (9+4)(25+16) \\ &= 13 \times 41\end{aligned}$$